

Q1 [40 points]: The capacitor shown in Fig. 1 consists of two parallel dielectric layers. Using the notion of electric field intensity (Coulomb's law), electric potential (boundary-value problem), boundary conditions for the surface separating the two dielectrics, obtain the expression of the overall capacitance.

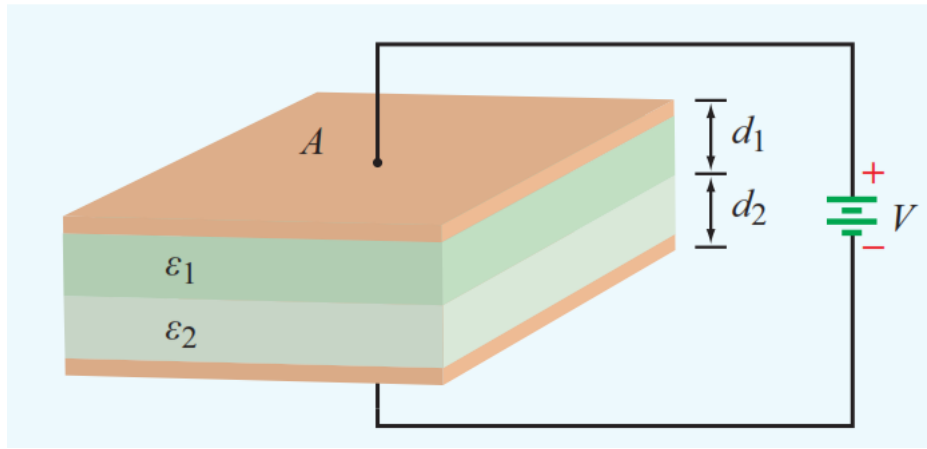


Figure 1: Capacitor with parallel dielectric layers .

Q2 [80 points]: An electron beam forms a current of density

$$\mathbf{J} = \begin{cases} J_0 \left(1 - \sqrt{\frac{\rho}{a}} \right) \mathbf{a}_z, & \rho < a \\ 0, & \rho > a \end{cases}$$

(a) Determine the total current.

(b) Using Ampere's circuit law, find the magnetic field intensity everywhere.

Q3 [60 points]: The two plates of the parallel-plate capacitor shown in Fig. 2 are each of length ℓ and width w , and the separation between them is d . The capacitor contains a dielectric block of dimensions $\ell \times w \times d$ and permittivity ϵ . The block can slide in and out of the capacitor cavity along its length dimension. You are required to compute the force $\mathbf{F}_e$ acting on the dielectric block when

it is partially outside of the cavity and the voltage across the capacitor is V . Steps are shown below:

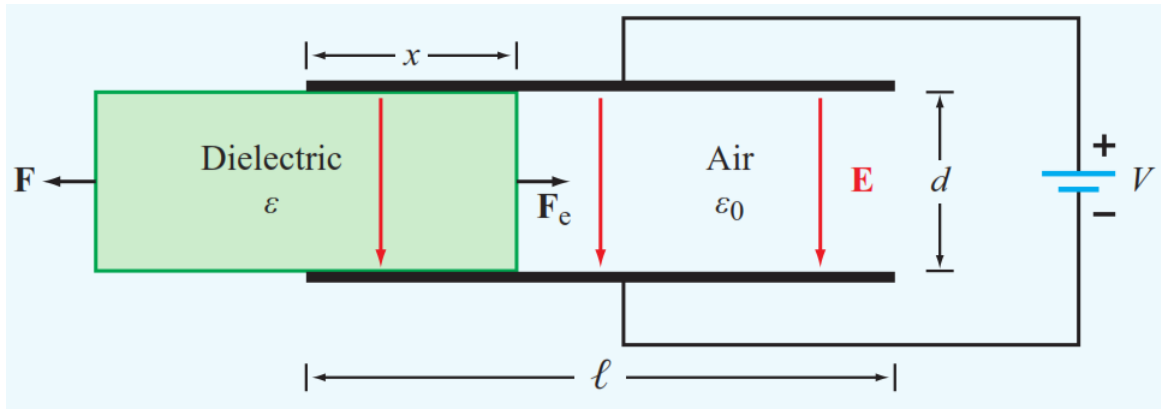


Figure 2: Parallel-plate capacitor with slidable dielectric block.

- (a) Compute the electric field intensity using any method of your choice. The width of the dielectric material inside the capacitor is x . Assume that there is no electric field in the other part of the dielectric material which is hanging outside. The fields in the air cavity and the material spaces differ due to difference in their permittivities; assume $\varepsilon \gg \varepsilon_0$.
- (b) Using electric field, evaluate the energies (W_E) in the dielectric and air regions; add them together to obtain the total energy.

$$W_E = \frac{\varepsilon_0}{2} \int_V |\mathbf{E}|^2 dv.$$

For the dielectric region, replace ε_0 with ε in W_E .

- (c) Under what scenario, W_E is maximum?
- (d) Sliding the dielectric block out of the capacitor requires exerting an external mechanical force $\mathbf{F}$ to oppose the electrostatic force $\mathbf{F}_E$, whose tendency is to oppose the reduction in W_E . Thus, the direction of $\mathbf{F}_E$ is to pull the block back into the capacitor. Compute $\mathbf{F}_E$ as follows:

$$F_E = \frac{dW_E}{dx}.$$

Good luck.

Q1 [40 points]: The capacitor shown in Fig. 1 consists of two parallel dielectric layers. Find the value of the overall capacitance.

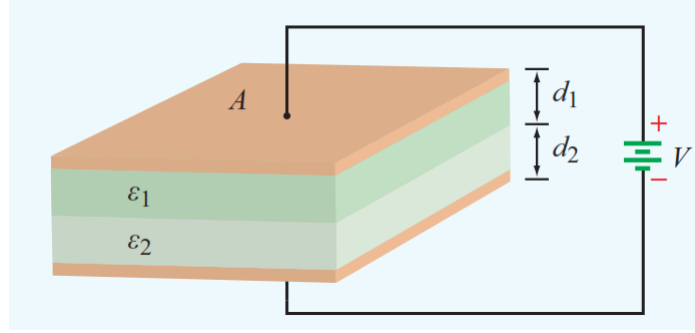


Figure 1: Capacitor with parallel dielectric layers .

Solution: Let the battery voltage be V_0 . The capacitance is obtained as:

$$C = \frac{Q}{V_0} = \frac{\int_S \epsilon \mathbf{E} \cdot d\mathbf{S}}{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}$$

Let $\mathbf{E}_1$ and $\mathbf{E}_2$ be the electric field in the cavities with permittivity, ϵ_1 and ϵ_2 , respectively. Since, the field are directed downward, from the top to the bottom plates, we may write:

$$\begin{aligned} \mathbf{E}_1 &= -E_1 \mathbf{a}_z, & d_2 < z \leq d_2 + d_1 \\ \mathbf{E}_2 &= -E_2 \mathbf{a}_z, & 0 \leq z < d_2. \end{aligned}$$

The charge Q may be obtained by taking a Gauss surface around the top plate (where the field is $\mathbf{E}_1$) as follows:

$$Q = \int_S \epsilon_1 \mathbf{E}_1 \cdot d\mathbf{S}$$

where $d\mathbf{S} = -dxdy\mathbf{a}_z$. Since the value of E_1 is same all over the plate, we obtain:

$$Q = \int_x \int_y \epsilon_1 E_1 dxdy = \epsilon_1 E_1 \int_x \int_y dxdy = \epsilon_1 E_1 A$$

Similarly, we obtain the line integral as follows:

$$V_0 = -\int_L \mathbf{E} \cdot d\boldsymbol{\ell}$$

where $d\boldsymbol{\ell} = dz\mathbf{a}_z$

$$V_0 = -\int_{z=0}^{d_2} \mathbf{E}_2 \cdot d\boldsymbol{\ell} - \int_{z=d_2}^{d_1+d_2} \mathbf{E}_1 \cdot d\boldsymbol{\ell} = E_2 d_2 + E_1 d_1$$

The capacitance is thus obtained as:

$$C = \frac{Q}{V_0} = \frac{\varepsilon_1 E_1 A}{E_1 d_1 + E_2 d_2} = \frac{\varepsilon_1 A}{d_1 + \frac{E_2}{E_1} d_2}$$

where all we need is to find the ratio of the fields E_1 and E_2 ; let us get help from the boundary conditions.

At $z = d_2$, the normal components satisfy $D_1 = \rho_S + D_2$, since $\rho_S = 0$, we get $\epsilon_1 E_1 = \epsilon_2 E_2$, or

$$\frac{E_2}{E_1} = \frac{\varepsilon_1}{\varepsilon_2}$$

Substituting this ratio in the expression of C , we obtain

$$\begin{aligned} C &= \frac{\varepsilon_1 A}{d_1 + \frac{\varepsilon_1}{\varepsilon_2} d_2} \\ &= \frac{\varepsilon_2 \varepsilon_1 A}{\varepsilon_2 d_1 + \varepsilon_1 d_2} \\ &= \frac{1}{\frac{\varepsilon_2 d_1}{\varepsilon_2 \varepsilon_1 A} + \frac{\varepsilon_1 d_2}{\varepsilon_2 \varepsilon_1 A}} \\ &= \frac{1}{\frac{d_1}{\varepsilon_1 A} + \frac{d_2}{\varepsilon_2 A}} \end{aligned}$$

Let

$$\begin{aligned} C_1 &:= \frac{\varepsilon_1 A}{d_1} \\ C_2 &:= \frac{\varepsilon_2 A}{d_2} \end{aligned}$$

We get

$$C = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2}} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} = C_1 \parallel C_2.$$

$$Q2. \quad \vec{J} = \begin{cases} J_0(1 - \sqrt{\rho/a}) \hat{z}, & \rho < a \\ 0, & \rho > a \end{cases}$$

(a)

$$I = \int \vec{J} \cdot d\vec{S}$$

$$d\vec{S} = \rho d\rho d\phi \hat{z}$$

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^a \vec{J} \cdot d\vec{S} \\ &= J_0 \int_0^{2\pi} \int_0^a \left(\rho - \frac{\rho^{3/2}}{\sqrt{a}} \right) d\phi d\rho \\ &= 2\pi J_0 \left(\frac{\rho^2}{2} - \frac{\rho^{5/2}}{5/2\sqrt{a}} \right) \Big|_0^a \\ &= 2\pi J_0 \left(\frac{a^2}{2} - \frac{2}{5} a^{5/2 - 1/2} \right) \\ &= 2\pi J_0 a^2 \left(\frac{1}{2} - \frac{2}{5} \right) \end{aligned}$$

$$\boxed{I_{total} = \frac{\pi}{5} J_0 a^2 \text{ [A]}}$$

(b) For $\rho < a$:

$$\oint \vec{H} \cdot d\vec{l} = I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

$$\text{LHS: } \oint \vec{H} \cdot d\vec{l} = 2\pi \rho H_\phi$$

$$\text{where } \vec{H} = H_\phi \hat{\phi}, d\vec{l} = \rho d\phi \hat{\phi}$$

$$\text{RHS: } \int \vec{J} \cdot d\vec{S} =$$

$$\begin{aligned} &\int_0^{2\pi} \int_0^\rho J_0 \left(\rho' - \frac{\rho'^{3/2}}{\sqrt{a}} \right) d\rho' d\phi \\ &= 2\pi J_0 \left(\frac{\rho^2}{2} - \frac{2\rho^{5/2}}{5\sqrt{a}} \right) \end{aligned}$$

Comparing both sides, we get

$$\boxed{H_\phi = J_0 \left(\frac{\rho}{2} - \frac{2\rho^{3/2}}{5\sqrt{a}} \right) \text{ [A/m]}}$$

(b) For $\rho > a$:

$$\text{LHS: } \oint \vec{H} \cdot d\vec{l} = 2\pi \rho H_\phi$$

$$\text{RHS: } \int_S \vec{J} \cdot d\vec{S} = I = \text{total current}$$

$$\Rightarrow \vec{H} = \frac{I}{2\pi \rho} \hat{\phi}$$

Note: From part (a), we have

$$I_{total} = \frac{\pi}{5} J_0 a^2$$

$$\Rightarrow \boxed{\vec{H} = \frac{J_0 a^2}{10\rho} \hat{\phi}}$$

Q3 [60 points]: The two plates of the parallel-plate capacitor shown in Fig. 2 are each of length ℓ and width w , and the separation between them is d . The capacitor contains a dielectric block of dimensions $\ell \times w \times d$ and permittivity ε . The block can slide in and out of the capacitor cavity along its length dimension. You are required to compute the force $\mathbf{F}_e$ acting on the dielectric block when it is partially outside of the cavity and the voltage across the capacitor is V . Steps are shown below:

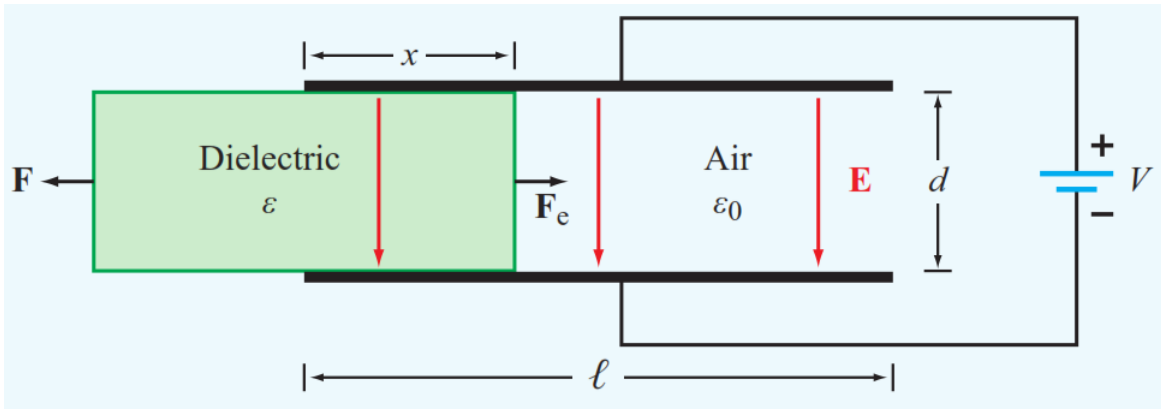


Figure 1: Parallel-plate capacitor with slidable dielectric block.

- Compute the electric field intensity using any method of your choice. The width of the dielectric material inside the capacitor is x . Assume that there is no electric field in the other part of the dielectric material which is hanging outside. The fields in the air cavity and the material spaces differ due to difference in their permittivities; assume $\varepsilon \gg \varepsilon_0$.
- Using electric field, evaluate the energies (W_E) in the dielectric and air regions; add them together to obtain the total energy.

$$W_E = \frac{\varepsilon_0}{2} \int_V |\mathbf{E}|^2 dv.$$

For the dielectric region, replace ε_0 with ε in W_E .

- Under what scenario, W_E is maximum?
- Sliding the dielectric block out of the capacitor requires exerting an external mechanical force $\mathbf{F}$ to oppose the electrostatic force $\mathbf{F}_E$, whose tendency is to oppose the reduction in W_E . Thus, the direction of $\mathbf{F}_E$ is to pull the block back into the capacitor. Compute $\mathbf{F}_E$ as follows:

$$F_E = \frac{dW_E}{dx}.$$

Solution: Let us use Laplace's equation to solving this problem. We have $\nabla^2 V = 0$. With two boundary conditions: $V = 0$ at $z = 0$; and $V = V_0$ at $z = d$; we need to solve:

$$V(z) = Az + B$$

Applying the boundary conditions, we obtain

$$V(z) = \frac{V_0}{d} z, \quad (A = V_0/d, B = 0)$$

Using, $V(z)$, we may obtain the electric field $\mathbf{E}$ as follows:

$$\mathbf{E} = -\nabla V = -\frac{\partial V}{\partial z} \mathbf{a}_z = -\frac{V_0}{d} \mathbf{a}_z.$$

The negative sign shows that the electric field is downward (which makes sense because the plate with high potential is above).

The capacitor has two regions; one is filled with dielectric of permittivity ε , and the other region is simply a free space with permittivity ε_0 ; note: $\varepsilon = \varepsilon_r \varepsilon_0 \gg \varepsilon_0$. Let us assume that the electric field in the two regions are $\mathbf{E}_1$ and $\mathbf{E}_2$ for dielectric and free space, respectively. From the above computation of Laplace's equation, we come to the conclusion that the field is same in both regions (dielectric and free space), i.e.,

$$\mathbf{E}_1 = \mathbf{E}_2 = -\frac{V_0}{d} \mathbf{a}_z.$$

So, we have $E^2 = |\mathbf{E}_1|^2 = |\mathbf{E}_2|^2 = V_0^2/d^2$. Apparently, this appears to be counter-intuitive; isn't it? You must be expecting that the field in the dielectric region must be much weaker than that in free space, since the field is known to be inversely proportional to permittivity. We discuss this issue separately. Given electric fields now known to us, we may evaluate the energies in the dielectric and air regions; the total energy.

$$\begin{aligned} W_E &= W_E^{\text{dielectric}} + W_E^{\text{air}} = \frac{\varepsilon}{2} \int_V |\mathbf{E}_1|^2 dv + \frac{\varepsilon_0}{2} \int_V |\mathbf{E}_2|^2 dv \\ &= \frac{\varepsilon}{2} E^2 \int_V^{\text{dielectric}} dv + \frac{\varepsilon_0}{2} E^2 \int_V^{\text{air}} dv \\ &= \frac{\varepsilon}{2} E^2 (d \times w \times x) + \frac{\varepsilon_0}{2} E^2 (d \times w \times (\ell - x)) \end{aligned}$$

W_E is maximum when the dielectric slab is completely inside, i.e., when $x = \ell$, giving

$$W_E^{\text{max}} = \frac{\varepsilon}{2} E^2 \ell w d.$$

The required force is obtained as

$$F_E = \frac{dW_E}{dx} = \frac{1}{2} E^2 w d (\varepsilon - \varepsilon_0) = \frac{1}{2d} V_0^2 w (\varepsilon - \varepsilon_0).$$

Discussion: We have just learned that the energy stored in a parallel capacitor is directly proportional to permittivity of the material filled between the plates. So the energy of the dielectric-capacitor is much higher than air-capacitor. Where this energy is coming from when the electric field is same in both dielectric and air capacitors (while being connected in parallel)?

Let us apply Gauss's theorem on the top plate around the dielectric region, we have

$$Q_{\text{enc}}^{\text{dielectric}} = \oint_S \mathbf{D}_1 \cdot d\mathbf{S} = \varepsilon \oint_S \mathbf{E}_1 \cdot d\mathbf{S}$$

Remember, all the flux which arises from the top positive plate flows downward and sinks into the lower negative plate. No flux is leaving in the upward direction, nor in the sideways. So we obtain

$$Q_{\text{enc}}^{\text{dielectric}} = \varepsilon \oint_S \mathbf{E}_1 \cdot d\mathbf{S} = \varepsilon Ewx = \varepsilon_0 \varepsilon_r Ewx$$

Similarly, we obtain

$$Q_{\text{enc}}^{\text{air}} = \varepsilon_0 \oint_S \mathbf{E}_2 \cdot d\mathbf{S} = \varepsilon_0 Ew(\ell - x)$$

Let us consider $x = \ell/2$, so that both capacitors have same width. We obtain

$$Q_{\text{enc}}^{\text{dielectric}} = \varepsilon_r Q_{\text{enc}}^{\text{air}} \gg Q_{\text{enc}}^{\text{air}}$$

assuming $\varepsilon_r \gg 1$. So, despite the fact that both capacitors have same electric field intensities (due to their parallel connection), the comparatively high energy in the dielectric-capacitor is due the large amount of charges it is holding compared to the air-capacitor.